Thesis sketches

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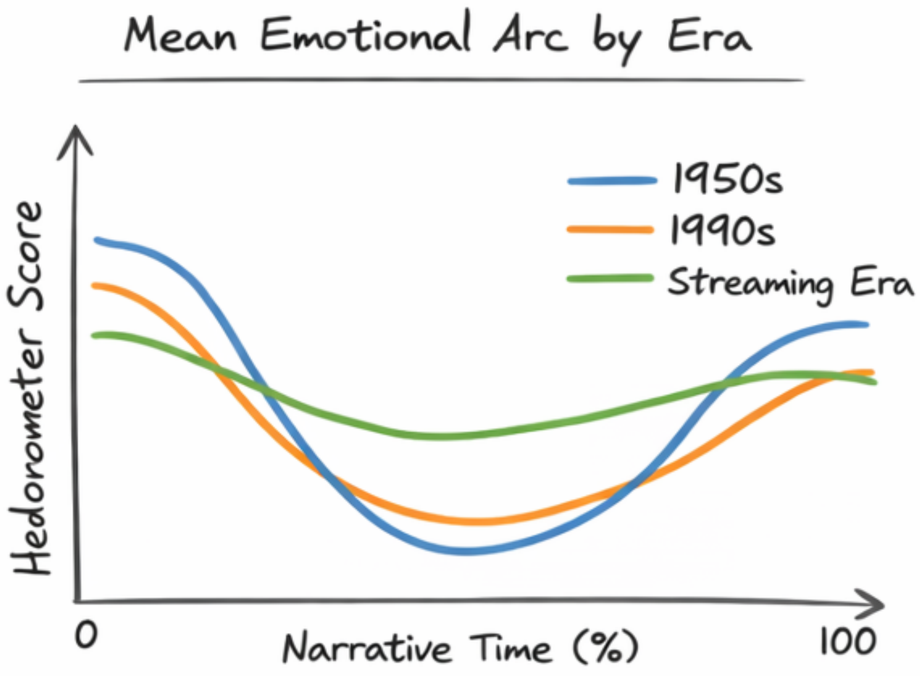
Emotional arc over time

- For **baseline shape comparison**
  - x-axis: normalised narrative time (0–100%)
  - y-axis: sentiment score (via wordlist for the Hedonometer)
- Possibly overlay: 1950s average arc, 1990s average arc, streaming era average arc
- Shows whether arc are flattening, fragmenting, compressing or intensifying
- If there is a drift, we should first see it here



Man in Hole

The main character gets into trouble then gets out of it again and ends up better off for the experience.

- Arsenic and Old Lace
- Harold & Kumar Go To White Castle

Boy Meets Girl

The main character comes across something wonderful, gets it, loses it, then gets it back forever.

- Jane Eyre
- Eternal Sunshine of the Spotless Mind

From Bad to Worse

The main character starts off poorly then gets continually worse with no hope for improvement.

- The Metamorphosis
- The Twilight Zone

Which Way Is Up?

The story has a lifelike ambiguity that keeps us from knowing if new developments are good or bad.

- Hamlet
- The Sopranos

Creation Story

In many cultures' creation stories, humankind receives incremental gifts from a deity. First major staples like the earth and sky, then smaller things like sparrows and cell phones. Not a common shape for Western stories, however.

Old Testament

Humankind receives incremental gifts from a deity, but is suddenly ousted from good standing in a fall of enormous proportions.

- Great Expectations

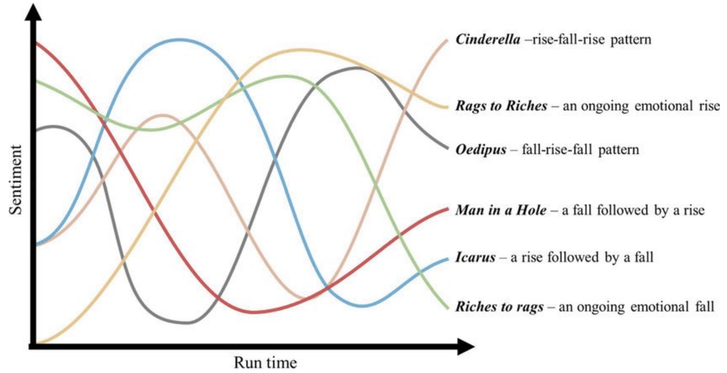
New Testament

Humankind receives incremental gifts from a deity, is suddenly ousted from good standing, but then receives off-the-charts bliss.

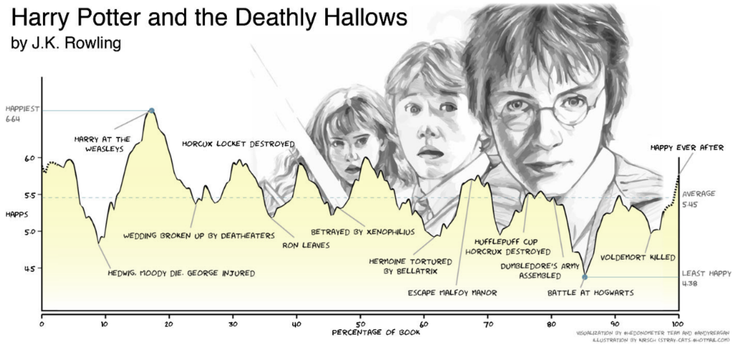
- Great Expectations with Dickens' alternate ending

Cinderella

It was the similarity between the shapes of Cinderella and the New Testament that thrilled Vonnegut for the first time in 1947 and then over the course of his life as he continued to write essays and give lectures on the shapes of stories.

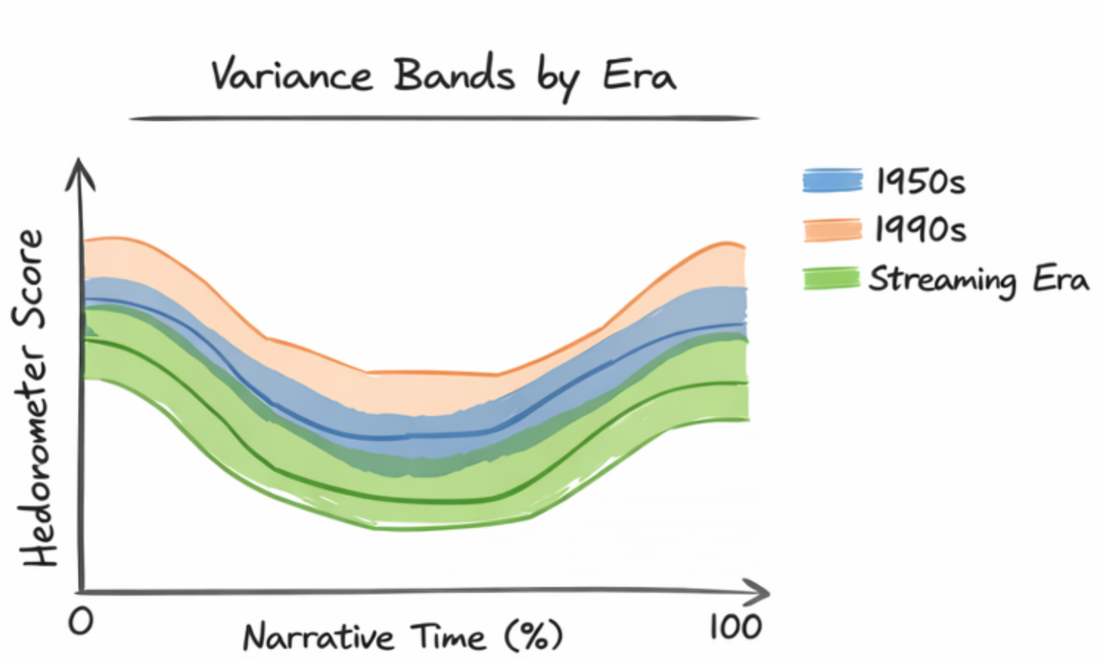


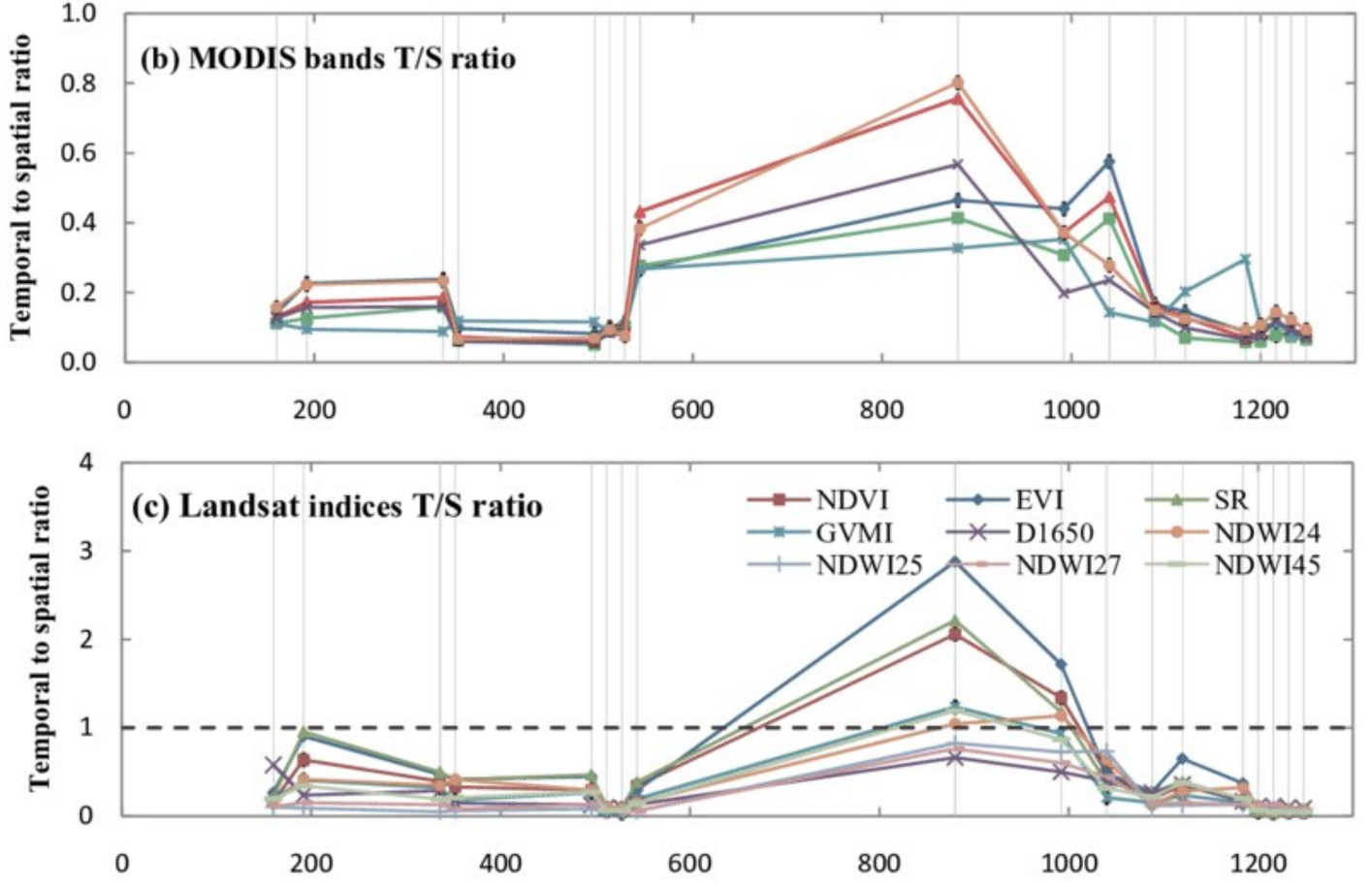
Harry Potter and the Deathly Hallows
by J.K. Rowling

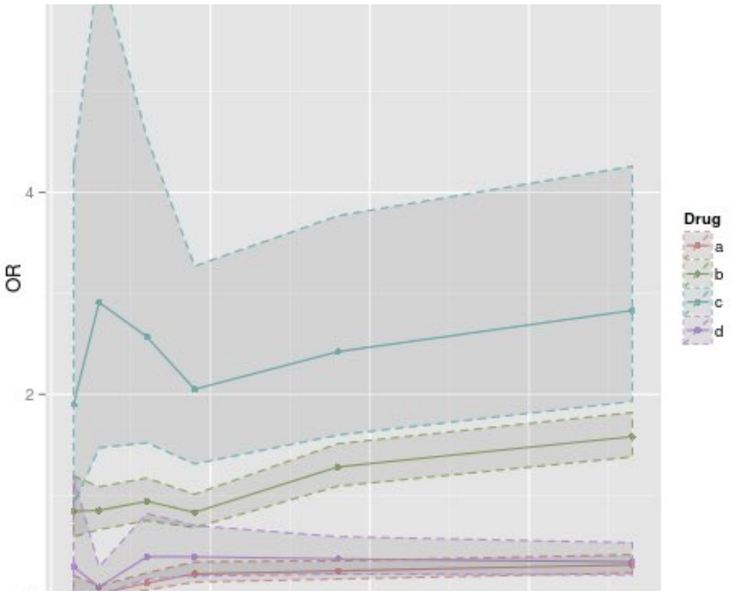


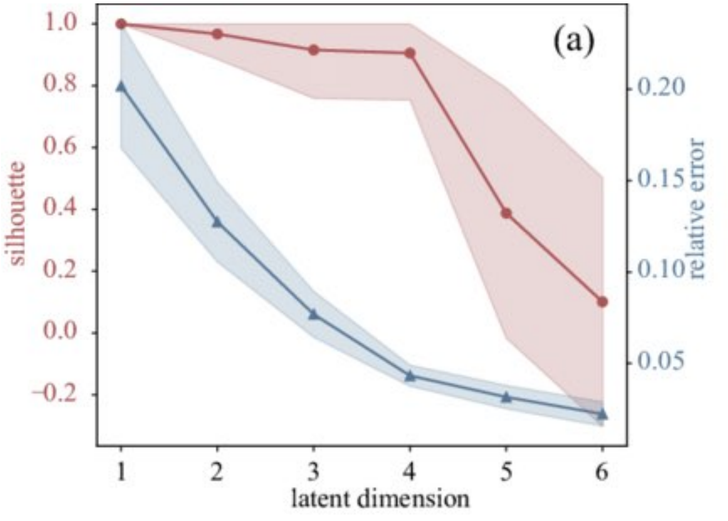
Variance bands

- For s**tability vs. chaos**
- Instead of just average arcs, show the spread
  - Plot mean line, add shaded confidence band, compare era-to-era variance
- If streaming-era narratives are structurally loser, the band widens
- If older films are formulaic, the band tightens
- Makes “structural drift” measurable



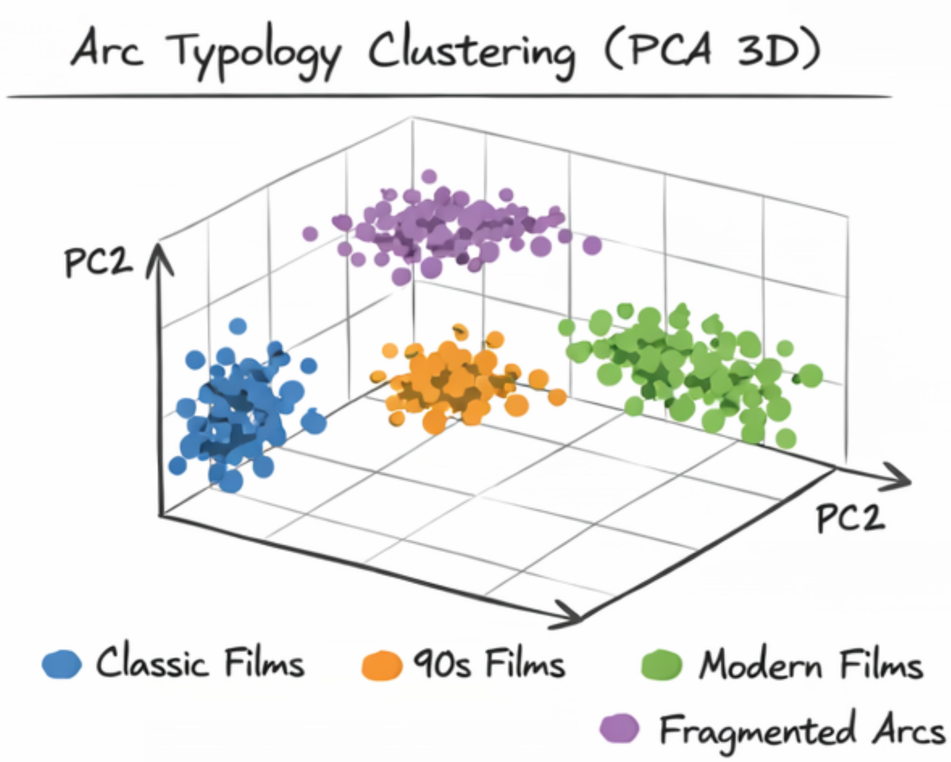

(b) MODIS bands T/S ratio
Temporal to spatial ratio
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
(c) Landsat indices T/S ratio
Temporal to spatial ratio
4
3
2
1
0
NDVI
EVI
SR
GVMI
D1650
NDWI24
NDWI25
NDWI27
NDWI45

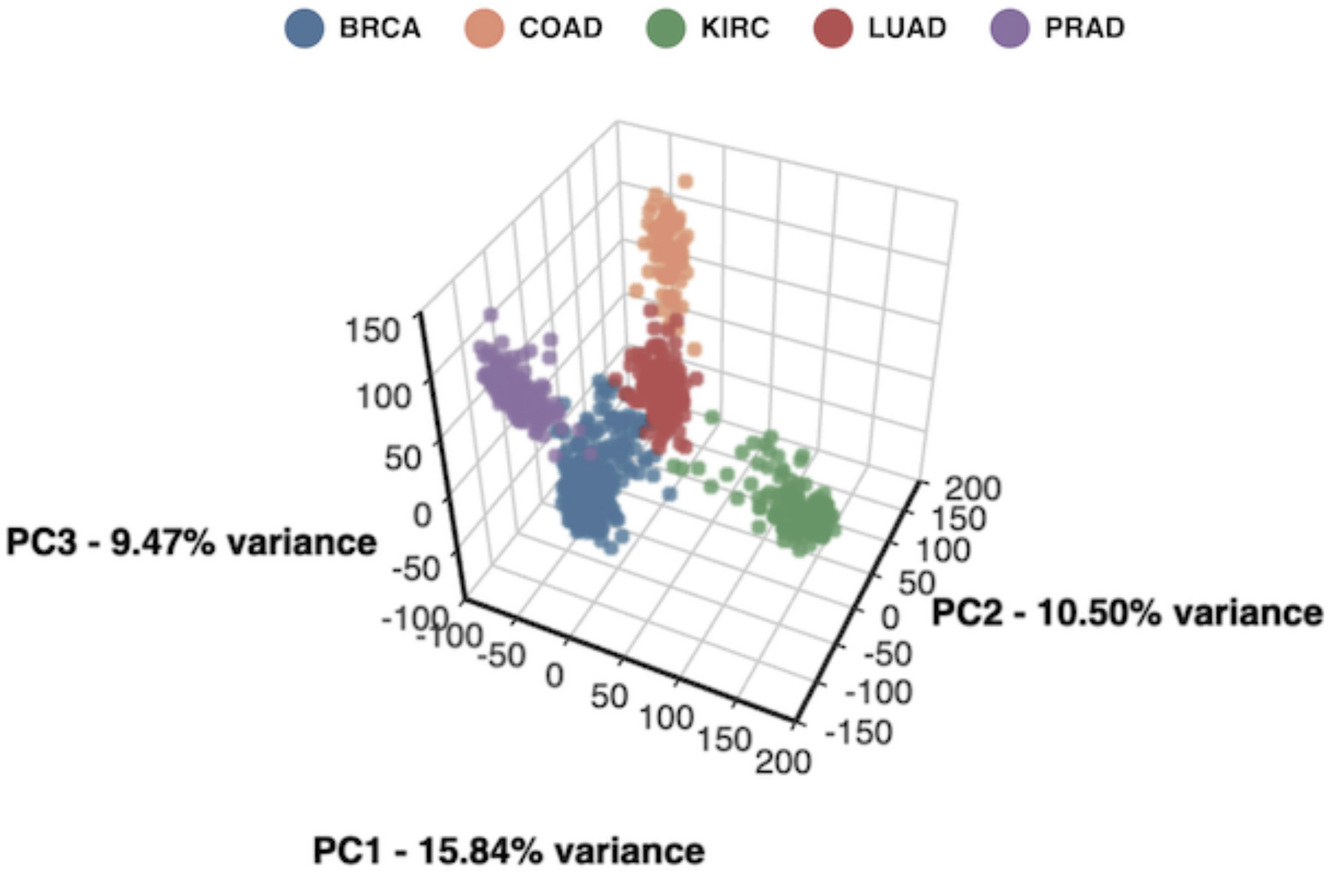

OR
4
2
Drug
a
b
c
d

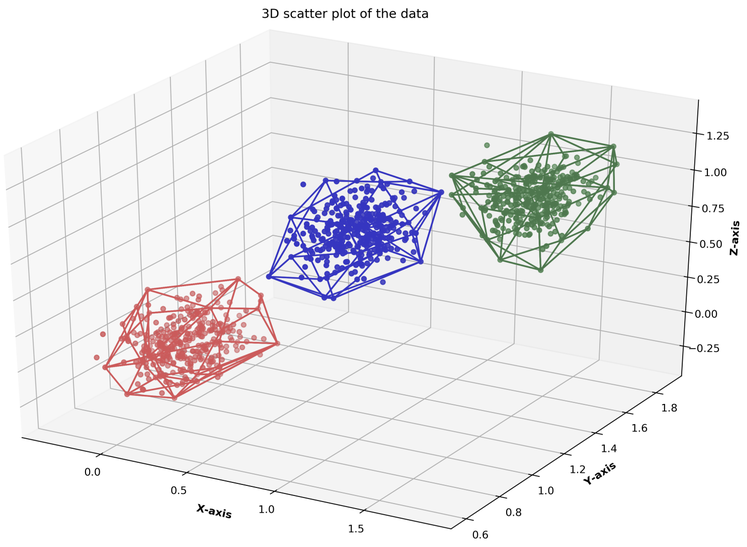

(a)
silhouette
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
relative error
0.20
0.15
0.10
0.05
1
2
3
4
5
6
latent dimension

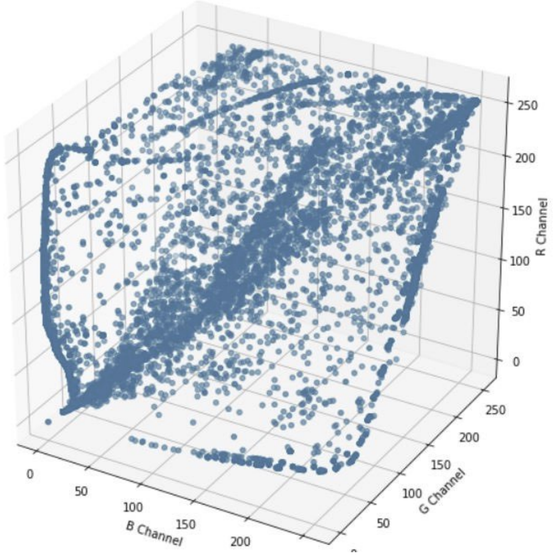
Arc typology clustering

- Are we telling different *types* of stories, or just reshaping the same ones?
- Treat each normalised arc as a 100-dimensional vector
- Then run k-means (k=4 or 5 to mirror Vonnegut's types), or hierarchical clustering, or time-series clustering (DTW, if serious)
- And then calculate distribution of cluster membership by era
- With PCA or UMAP, each dot is one film and colour is era
- If eras cluster separately in shape-space, that is a structural drift




BRCA
COAD
KIRC
LUAD
PRAD
PC3 - 9.47% variance
PC2 - 10.50% variance
PC1 - 15.84% variance


3D scatter plot of the data
X-axis
Y-axis
Z-axis


R Channel
G Channel
B Channel

I have more ideas for different visualisations
but these should do for now